

Aman Singh

+91 7054470414 | saman@iisc.ac.in | [LinkedIn](#) | [Website](#) | [Google Scholar](#)

EDUCATION

Indian Institute of Science

Doctor of Philosophy in Cyber-Physical Systems

Bengaluru, India

August 2022 – Present

Indian Institute of Technology Roorkee

Bachelor of Technology in Mechanical Engineering

Roorkee, India

July 2015 – May 2019

PROFESSIONAL EXPERIENCE

Project Associate

RBCCPS, Indian Institute of Science

Jan 2021 – July 2022

Bengaluru, India

- Developed a high torque density actuator with applications in robots like biped, quadruped etc
- Developed a chain-sprocket based 2R manipulator leg using the above actuators
- Developed a 12 DOF highly dynamic quadruped robot named Stoch-3

Verification Engineer

Oski Technology

July 2019 – Jan 2021

Gurugram, India

- Signed off Register Transfer Level designs of integrated circuits using formal verification
- Verified an AXI interconnect connecting 10 AXI masters and 18 AXI slaves
- Verified the resource management functionality of a block that receives custom instructions from 2 RISC-V cores

SELECTED PUBLICATIONS

COMPAct: Computational Opt. & Automated Modular Design of Planetary Act. ICRA 2026

A. Singh, D. Kapa, S. Joshi, and S. Kolathaya | [Paper-Link](#) | [Video-Link](#)

Accepted

A Co-Design FW for Energy-Aware Monoped Jumping, Detailed Act. Model Humanoids 2025

A. Singh, A. Mishra, D. Kapa, S. Joshi and S. Kolathaya | [Paper-Link](#) | [Video-Link](#)

Published

Comparison between ESSPG & ISSPG Actuators for Legged Robots

AIR 2025

A. Singh, D. Kapa, P. Chedda, and S. N. Y. Kolathaya | [Paper-Link](#)

Accepted

A Chain-Driven, Sandwich-Legged Quadruped Robot: Design and Exp. Analysis

AIR 2025

A. Singh, B. G. Goswami, K. Nehete, and S. N. Y. Kolathaya | [Paper-Link](#) | [Video-Link](#)

Accepted

PATENTS

Low-cost sandwiched robotic leg design for legged locomotion

Aman Singh, Shishir Kolathaya | [Patent-certificate](#)

Indian Patent Office (IPO)

Accepted

PROJECTS AND COMPETITIONS

Human-Inspired Biped Walking | [Video](#) | Built a 2D biped test-bed & walking controller Aug 2018 – Apr 2019

Compliant Robot Leg, Polytechnique Montreal | Designed & built a compliant leg May 2018 – Jul 2018

ABU Robocon 2018, Team Leader | [Video](#) | 7th/109 teams, Best Innovative Design award Jun 2017 – Mar 2018

ABU Robocon 2017, Technical Member | Built a mecanum-drive frisbee-throwing robot Jun 2016 – Mar 2017

TECHNICAL SKILLS

Software Tools: MATLAB, SolidWorks, Ansys, ROS, Gazebo, PyBullet

Programming Languages: C++, Python, LaTeX

HONOURS AND AWARDS

MITACS Globalink Award, 2018: MITACS

OPJEMS Scholarship, 2016: O. P. Jindal Group

- COMPAct: Computational Opt. & Automated Modular Design of Planetary Act.** ICRA 2026
A. Singh, D. Kapa, S. Joshi, and S. Kolathaya | [Paper-Link](#) | [Video-Link](#) *Accepted*
- A Co-Design FW for High-Perf. Jumping of a Five-Bar Monoped w/ Act. Opt.** AIM 2026
A. Mishra, A. Singh, and S. Kolathaya | [Paper-Link](#) *Accepted*
- DTEA: A Dual-Topology Elastic Actuator w/ Series-Parallel Switching** IROS 2026
V. Ramesh, A. Singh, and S. Kolathaya | [Paper-Link](#) *Accepted*
- A Co-Design FW for Energy-Aware Monoped Jumping, Detailed Act. Model** Humanoids 2025
A. Singh, A. Mishra, D. Kapa, S. Joshi and S. Kolathaya | [Paper-Link](#) | [Video-Link](#) *Published*
- Comparison between ESSPG & ISSPG Actuators for Legged Robots** AIR 2025
A. Singh, D. Kapa, P. Chedda, and S. N. Y. Kolathaya | [Paper-Link](#) *Accepted*
- A Chain-Driven, Sandwich-Legged Quadruped Robot: Design and Exp. Analysis** AIR 2025
A. Singh, B. G. Goswami, K. Nehete, and S. N. Y. Kolathaya | [Paper-Link](#) | [Video-Link](#) *Accepted*
- Force Control for Robust Quadruped Locomotion: A Linear Policy Approach** ICRA 2023
A. Shirwatkar, V. K. Kurva, D. Vinoda, A. Singh, et al. | [Paper-Link](#) *Published*
- Dynamic Mirror Descent based MPC for Accelerating Robot Learning** ICRA 2022
U. A. Mishra, S. R. Samineni, P. Goel, C. Kunjeti, H. Lodha, A. Singh, et al. | [Paper-Link](#) *Published*